SB1 · STATIC-BEAM

Euler-Bernoulli 1D cantilever (tip deflection)

Tip deflection under a transverse point load — 1-D flexural absolute check

PASS

error 0.00% · tol 1%

§2 TECHNICAL FEATURES TESTED

- **STRIX engine 2-node frame element** — linear static
- Euler–Bernoulli flexure (no shear area emitted)
- Strong-axis routing **$Iy ← sec.Iz**
- Linear coordinate transformation, auto local-z reference vector [0,0,1]
- Nodal load & single-node fixity

§3 PROBLEM DESCRIPTION

A prismatic cantilever of length L is fully fixed at node N1 and free at node N2, where a downward transverse load P acts along global −Z. The tip deflection is compared with the Euler–Bernoulli closed-form solution. Because the static element carries no shear area, the response is pure bending — the element should reproduce the closed form to machine precision, independent of subdivision.

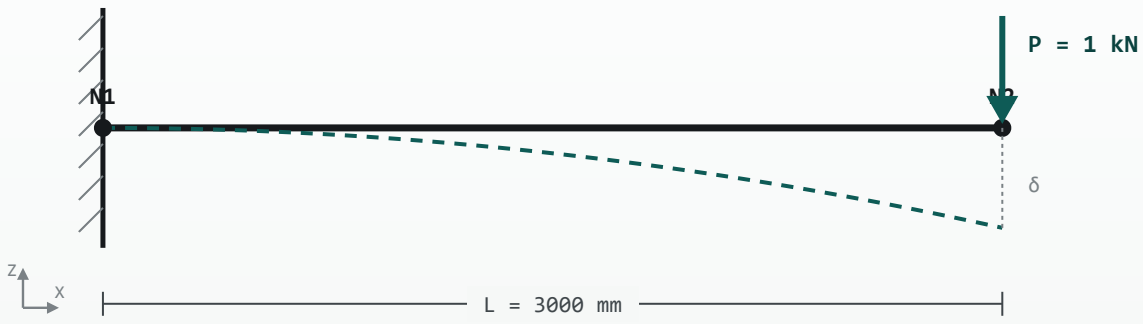


Figure SB1. Cantilever geometry, loading, and deflected shape (deflection exaggerated).

§4 GEOMETRY, PROPERTIES & LOADING

GEOMETRY		MATERIAL		LOADING & BC	
Length L	3000 mm	Model	Elastic	N1	fixed 6-DOF
Section b×h	300×500	E	26700 MPa	N2 load	Fz −1000 N
Area A	150000	ν	0.2	Self-weight	off
I strong (Iz)	3.125×10^{9}	Units	N · mm	Analysis	Linear static
I weak (Iy)	1.125×10^{9}				

§5 REFERENCE SOLUTION

Tip deflection u_z: -P*L^3/(3*E*I) = −0.107865 mm

Tip rotation r_y: +P*L^2/(2*E*I) = 5.3933×10^{-5} rad

Timoshenko & Gere, Mechanics of Materials — cantilever tip deflection under an end load (Euler-Bernoulli).

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Tip deflection u_z (mm)	N2.uz	−0.107865	−0.107865	0.00%	PASS
Tip rotation r_y (rad)	N2.ry	5.3933×10^{-5}	5.3933×10^{-5}	0.00%	PASS
Support reaction R_z (N)	N1.Fz	1000	1000	0.00%	PASS
Support moment M_y (N·mm)	N1.My	-3×10^{6}	-3×10^{6}	0.00%	PASS

§7 CONVERGENCE

MESH	UZ (MM)	DIFF
1 element	−0.107865	0.00%

The cubic Hermite shape functions of the Euler–Bernoulli element are exact for a linearly varying moment field, so a single element already recovers the closed form. Subdivision is `mesh-independent` — no refinement study is required for this benchmark.

§8 CONCLUSION

PASS STRIX reproduces the reference solution to **0.00%** (tolerance 1%). The tested element and its formulation are verified against the reference.

§9 REFERENCES & ANALYSIS FILES

1. Timoshenko, S. & Gere, J. — *Mechanics of Materials.* Cantilever end-load deflection $\delta = PL^3/3EI$.

2. STRIX headless harness: `generatePyForLoadCase → STRIX engine (static-single route) → node_disp.out`.

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · SB1/` `sha256 (pending publish)`